

Tool_Box v1.0

Full Functionality Guide

Menu-driven Linux / network lab toolbox

Authorized use only

Use Tool_Box only on systems, networks, services, accounts, and data that you own or are explicitly authorized to test. Tool_Box coordinates third-party utilities; those tools retain their own behavior, risks, and licensing.

This guide documents the v1.0 interface, workflows, menus, output handling, settings, external-module behavior, and operational conventions. It intentionally does **not** teach module creation; see **Tool_Box_External_Module_Guide.pdf** for that separate documentation.

Repository documentation filenames

Tool_Box_Functionality_Guide.pdf
Tool_Box_Quick_Guide.txt
Tool_Box_External_Module_Guide.pdf

Version 1.0

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1. What Tool_Box Is

Tool_Box is a Bash-based menu wrapper for common Linux, networking, reconnaissance, enumeration, traffic-analysis, reporting, and lab-security utilities. It keeps command construction visible instead of hiding what will be executed, and it adds shared target data, output organization, workspaces, command history, documentation shortcuts, software checks, and optional external modules.

- **Command visibility:** command pages show a preview before execution.
- **Consistent execution:** command pages use **R** as the primary Run key.
- **Global navigation:** the footer provides Back, Main Menu, Set Target/Data, Workspace, and Help shortcuts.
- **Dry-Run:** preview supported actions without executing them.
- **Auto-Save:** capture command output into the configured results tree.
- **Workspaces:** keep target state, notes, findings, and imported services together.
- **External modules:** add menu commands from reviewed .txt definitions without editing the main script.

AI-assisted development

Tool_Box has been developed with substantial AI assistance ("vibe coded"). Review the source, validate command previews, and test new functionality in an isolated lab before relying on it.

1.1 Repository layout

```
Tool_Box/  
|-- ToolBox.sh  
|-- README.md  
|-- LICENSE  
|-- TERMS.md  
|-- Tool_Box_Functionality_Guide.pdf  
|-- Tool_Box_Quick_Guide.txt  
|-- Tool_Box_External_Module_Guide.pdf  
`-- Toolboxmodules/  
    |-- Basic_curl.txt  
    `-- Basic_nmap.txt
```

GitHub-facing documentation uses stable filenames without version numbers. Release archives can still use versioned names such as **Tool_Box_v1.0.zip**.

2. Starting Tool_Box

Launch directly with Bash or make the script executable:

```
bash ToolBox.sh  
  
# or  
chmod +x ToolBox.sh  
./ToolBox.sh
```

Tool_Box locates its **Toolboxmodules** directory relative to the script itself, so the working directory you launch from does not need to be the repository directory.

2.1 Startup splash and terms acknowledgement

Every normal launch shows a short ASCII-art splash. The first launch of each Tool_Box version for a Linux user also displays the Terms & Authorized-Use Notice.

Accept terms and continue? [Y/n]:

- Press **Enter**, **Y**, or **y** to accept and continue.
- Enter **N** or **n** to exit without loading the main menu.
- Acceptance is recorded in `~/.tool_box_terms.conf`.
- A new Tool_Box version asks for acknowledgement again, even if the terms revision has not changed.
- Removing the acceptance file causes the notice to appear again.

Important

The acknowledgement is a project usage notice, not a technical security boundary and not a substitute for understanding the commands you run.

3. Interface and Navigation

The header shows the current target, subnet, selected ports, output folder, active workspace, current modes, and whether Tool_Box is running as root or a standard user.

3.1 Global footer

[0] Back [M] Main Menu [T] Set Target/Data [W] Workspace [H] Help

- **0** - return to the previous local menu.
- **M** - restart directly at the main menu while carrying forward the active session state.
- **T** - open Set Target/Data, then return to the main menu.
- **W** - open Workspace / Project, then return to the main menu.
- **H** - open Quick Help / Workflow Guide.

Non-main menus do not repeat a separate **0) Back** item because Back is already shown in the footer.

3.2 Command-page layout

Most command builders use consistent visual sections:

- **Command Preview** - the resolved command Tool_Box is preparing.
- **Options** - settings and toggles.
- **Customization** - module-local placeholder overrides or custom arguments where available.
- **Actions** - usually **R) Run**.

- **Documentation** - man page, help, or module help.

3.3 Run, confirmation, privilege, and Dry-Run behavior

Commands are shown again before execution. Elevated actions receive an extra privilege warning. Dry-Run records a preview without executing the command. Auto-Save, when enabled, writes supported command output to the configured result tree.

4. Main Menu Map

1) Target & Scanning

- Workspace / Project
- Set Target / Data
- Scan Profiles
- Import Nmap Results
- Service-Aware Recommendations

2) Reconnaissance & Enumeration

- Reconnaissance
- Web Enumeration
- Network / DNS
- SMB / Windows
- Service Enumeration
- Vulnerability Assessment

3) Access & Exploitation

- Pass / Hash Cracking
- Exploit

4) Analysis & Reporting

- Traffic Analysis
- Results / Reporting

5) Utilities & Resources

- Command-Line Utilities
- Network Information / VPN Status
- Text / Encoding Tools
- Port Reference
- Useful Links
- Quick Help / Workflow Guide

6) System & Configuration

- Install / Manage Software
- Settings

- Tool_Box Diagnostics
- Privilege / User Session
- About / Version / Update Check
- Documentation / Download Guides

5. Target & Scanning

This area controls project state and the target information used by most other menus.

5.1 Workspace / Project

Workspaces keep project metadata separate from raw command output. The workspace menu can create, load, save, archive, unload, list, or delete workspaces.

- **Workspace Dashboard** - shows the active target, selected ports, result directory, result-file count/size, imported services, note count, and finding count.
- **Notes** - add, view, or edit Markdown notes.
- **Findings / Loot Tracker** - add, view, and search categorized findings such as host, service, username, credential, URL, file, hash, or other.
- **Import Nmap Results** - parse Nmap XML, grepable output, or normal text into a workspace service inventory. Import can update the current target and selected ports.
- **Service-Aware Recommendations** - maps imported/selected services to relevant Tool_Box areas; it does not automatically launch scans.

5.2 Set Target / Data

Shared session values include IP, subnet, port selection, output folder, domain, URL, username, and password. These values feed built-in tools and external-module placeholders.

Password handling

The shared password is treated as session-sensitive data. Tool_Box redacts supported password values from its own previews/history, but a third-party program may still display or log them.

5.3 Scan Profiles

- **Quick Recon** - Nmap fast scan with service detection.
- **Full TCP Recon** - scans all TCP ports with service detection.
- **Web Server** - combines Nmap, WhatWeb, Curl headers, and optional Gobuster content discovery.
- **Windows / SMB** - combines Nmap SMB information scripts and anonymous SMB share listing when tools are available.
- **Linux Server** - service/default-script Nmap scan on selected ports.
- **Custom** - opens the Nmap builder.

- **Scan Profiles Software** - checks or manages profile-related software.

5.4 Import Nmap Results and recommendations

Nmap import accepts XML, .gnmap, and normal text output. The parser extracts open services into the workspace service inventory. The recommendation screen can then point you toward Web Enumeration, SMB / Windows, Network / DNS, or Service Enumeration based on ports and detected services.

6. Reconnaissance & Enumeration

This is the largest operational area. It groups target discovery, web checks, DNS/network inspection, SMB/Windows enumeration, service-specific checks, and vulnerability-oriented assessments.

6.1 Reconnaissance

- **Nmap - Custom Builder** - choose SYN, Connect, or UDP scan type; toggle version detection, OS detection, default scripts, skip-ping, T4 timing, aggressive scan, custom options, and global Auto-Save. **R** runs the command and **D** opens Nmap help.
- **Host Discovery** - Nmap-based discovery workflow for the selected target/subnet.
- **Port Discovery** - port-oriented Nmap discovery using the current target.
- **Service Detection** - Nmap service/version detection on selected ports.
- **DNS Enumeration** - opens DNS lookup functionality.
- **WHOIS** - accepts a custom IP/domain or the global target.
- **Traceroute** - accepts a custom target and can reset to the global target.
- **Ping** - temporary IP/domain override, request count, timeout, numeric output, reset, Run, and help.
- **Reconnaissance Software** - software status/management for relevant tools.

6.2 Web Enumeration

Tool / Page	Purpose
Gobuster	Wordlist-driven web content discovery with configurable protocol, port, wordlist, threads, extensions, and custom options.
FFUF	Fast web fuzzing with protocol, port, wordlist, threads, extensions, and custom options.
Feroxbuster	Recursive content discovery with similar target/wordlist/thread controls.
WhatWeb	Web technology fingerprinting on a selected protocol/port.
Nikto	Web server checks for common security issues and risky content.
HTTP Headers (Curl)	Fetch response headers for the selected web endpoint.
TLS Certificate (OpenSSL)	Inspect a TLS endpoint with configurable port and server name.
Test HTTP/HTTPS	Simple HTTP/HTTPS connectivity check.

6.3 Network / DNS

- **DNS Lookup** - query DNS records with Dig.
- **Reverse DNS** - PTR-style lookup for a custom IP.
- **WHOIS** and **Traceroute** - target/domain information and route inspection.
- **ARP Scan** - local-network ARP discovery; may require elevated privileges.
- **Test TCP Port (Netcat)** - check connectivity to a target/port.
- **Local Interfaces** and **Routing Table** - inspect local network state.

6.4 SMB / Windows

- **SMB Enumeration (enum4linux-ng)** - gather SMB/Windows host information.
- **List SMB Shares** - anonymous share listing with smbclient.
- **SMB Client** - connect to a selected share with optional username.
- **Nmap SMB Information Scripts** - run SMB-focused information scripts.
- **NetBIOS Information** - collect NetBIOS-oriented information via Nmap.

6.5 Service Enumeration

Service	Available workflow
SSH	Set port; Nmap version detection; SSH host-key scan; Nmap SSH algorithms/host keys; SSH/Nmap help.
FTP	Set port; banner/version checks; anonymous/system-information Nmap checks.
SMTP	Set port; banner; SMTP command enumeration; service detection.
SNMP	Set port, community string, and OID; run snmpwalk; community string is redacted from Tool_Box history.
LDAP	Set port/base DN; query RootDSE or selected base DN.
NFS	List exports; show RPC services.
RPC	List registered RPC programs.
Database Services	Set database ports and use Nmap detection for common database services.

6.6 Vulnerability Assessment

This menu provides selected Nmap/NSE-based assessment checks for systems you are authorized to test. It includes an assessment port setting, safe NSE script scan, TLS cipher assessment, HTTP security-header check, SMB security-configuration check, combined service detection/default scripts, software status, and Nmap documentation.

Assessment is not a verdict

A successful command or a clean-looking result does not prove a system is secure. Treat results as evidence to review, not as an automatic vulnerability determination.

7. Access & Exploitation

This top-level area currently contains Pass / Hash Cracking helpers and Metasploit access.

7.1 Pass / Hash Cracking

The current menu provides a hash-identification helper and access to Text / Encoding Tools. It does not automatically recover credentials by itself; it helps identify/prepare data and route you to supporting utilities.

7.2 Exploit / Metasploit

The Exploit menu currently exposes Metasploit. The Metasploit page provides **R) Launch msfconsole**, **V) Show Version**, and **D) View Help**. Use only in environments where you have explicit permission.

8. Analysis & Reporting

8.1 Traffic Analysis

- **Start tcpdump Capture** - choose interface and packet count, then capture traffic.
- **Read Capture** - select a PCAP file and read it with tcpdump.
- **List Capture Interfaces** - show tcpdump capture interfaces.
- **Capture by Host** and **Capture by Port** - focused capture helpers.
- **TShark Analysis** - select a PCAP and optional display filter.
- **Traffic Analysis Software** - status/management for tcpdump and tshark.

Privilege note

Live packet capture commonly needs root privileges or Linux capabilities. Tool_Box flags elevated actions and asks for confirmation.

8.2 Results / Reporting

- **Results Dashboard** - current target, result file count, category count, size, command-history count, and workspace counts when available.
- **List Targets with Results** and **List Recent Result Files** - browse stored output.
- **View a Text Result** - inspect saved text output.
- **Search Current Target Results** - recursive text search; also searches workspace notes/findings when a workspace is active.

- **Compare Two Text Results** - compare saved text outputs.
- **Generate Markdown Target Report** - build a Markdown summary from target results/workspace information.
- **Archive Current Target Results** - package current target output.
- **Delete a Result File** - destructive action with confirmation.
- **Command History** - view or search Tool_Box command/result history.

9. Utilities & Resources

9.1 Command-Line Utilities

Utility	What the menu exposes
Curl	Set URL, GET request, headers-only request, and help.
Wget	Set URL and download.
Netcat TCP Test	Connectivity test helper.
JQ	Set JSON file and pretty-print JSON.
OpenSSL	Show version, SHA-256 hash a file, TLS certificate viewer, and help.

9.2 Network Information / VPN Status

- Local interfaces / routes / DNS.
- VPN / tunnel status.
- Public IP lookup using an external request.

9.3 Text / Encoding Tools

- Base64 encode/decode.
- Hex encode/decode.
- URL encode/decode.
- SHA-256 text hashing.
- Hash identification helper.

9.4 Port Reference

Provides a quick reference for common service ports from inside Tool_Box.

9.5 Useful Links

Useful Links groups browser-accessible references into the following categories:

- **Shell / Session Tools** - Penelope, Reverse Shell Generator.
- **Linux / Windows Privilege References** - GTF0Bins, LOLBAS, HackTricks.

- **Data / Encoding / Analysis** - CyberChef, VirusTotal.
- **Web Security** - PortSwigger Web Security Academy, OWASP Cheat Sheet Series, OWASP Web Security Testing Guide.
- **Recon / Vulnerability References** - crt.sh, Nmap reference/NSE docs, Exploit Database, MITRE ATT&CK, CVE.org, NIST NVD.
- **Payloads / Wordlists / Cheat Sheets** - PayloadsAllTheThings, SecLists, PacketLife Cheat Sheets.
- **Training** - TryHackMe and PortSwigger Web Security Academy.
- **Custom Bookmarks** - user-managed saved bookmarks.

9.6 Quick Help / Workflow Guide

The built-in Quick Help suggests a workflow: create/load a workspace, set target data, run Nmap or a profile, import Nmap results, review service-aware recommendations, record notes/findings, then use Results / Reporting to search output and generate a report.

10. System & Configuration

10.1 Install / Manage Software

- **Scan for Installed** - refresh supported-tool installation state.
- **Install / Manage Supported Software** - individual tool status, install/repair, version, help, and repository troubleshooting.
- **Install Missing** - install supported missing packages on compatible apt-based systems.
- **Repository Troubleshooter** - inspect OS/repositories, enable standard repositories for the detected OS, update package lists, check package availability, and run APT diagnostics.
- **Show Supported Software Versions / Show Missing Supported Software** - inventory views.

Tool_Box detects the operating system and is designed not to mix Kali repositories into Ubuntu/Debian or vice versa.

10.2 Settings

- Default wordlist and default threads.
- Output directory.
- Auto-Save Output, Verbose Mode, Dry-Run Mode, and Color Output toggles.
- Save / Load Configuration and Reset Defaults.
- Settings Profiles - save, load, delete, or list reusable settings profiles.
- External Modules - list, load, unload, and rescan module definitions.

10.3 Diagnostics

- **R) Run All Checks.**

- Software status.
- Output folder check.
- Default wordlist check.
- Network interfaces/routes.
- Configuration check.
- Privilege/package-manager information.
- Bash syntax check.
- APT/repository troubleshooter.
- Desktop/browser/clipboard check.

10.4 Privilege / User Session

A standard-user session can start a nested root Tool_Box session. The current target, ports, output path, and active settings are carried into the elevated child. Exiting the root child returns to the original standard-user Tool_Box session.

10.5 About / Version / Update Check

- Open or copy the repository URL.
- Check repository version.
- View the privilege guide.
- Download documentation.
- View the Terms / Authorized-Use notice and acceptance state.

10.6 Documentation / Download Guides

Tool_Box expects these stable repository documentation names:

```
Tool_Box_Functionality_Guide.pdf
Tool_Box_Quick_Guide.txt
```

The menu can download the full PDF guide, the quick TXT guide, or both; set a download location; or open the repository.

11. External Modules - How They Work

External modules let Tool_Box load additional command pages from reviewed **.txt** definitions in the **Toolboxmodules** directory. They are treated as data rather than sourced as Bash code. Loading a definition does not run its command.

11.1 Module manager

Open **System & Configuration -> Settings -> External Modules**. The manager can list discovered/selected modules, load one, load all, unload one, unload all, and rescan/reload selected definitions.

Loaded modules appear in their configured menu category with selectors such as **E1**, **E2**, and so on. Module numbering is shared across categories, so gaps in a particular menu are normal.

11.2 Module command pages

A module can provide up to three independent option groups. Version 1.0 uses reserved structural markers **<<OPTIONS>>**, **<<OPTIONS2>>**, and **<<OPTIONS3>>** so enabled option fragments can be inserted at specific locations in the final command. Existing legacy modules that use only an **options:** section without the first marker keep append-at-end behavior.

- **C - Custom Placeholder Values:** temporarily override normal values such as IP, URL, port, wordlist, username, or other placeholders for this module only.
- **A - Custom Arguments:** append temporary validated arguments.
- **X - Clear Customizations:** clear module-local overrides and custom arguments.
- **R - Run:** execute through the standard Tool_Box confirmation/logging flow.
- **D - Module Help:** explain option groups and module controls.

Separate creation documentation

This full functionality guide intentionally does not include instructions for writing module files. For module creation syntax, category hierarchy/aliases, examples, validation rules, templates, and troubleshooting, use **Tool_Box_External_Module_Guide.pdf**.

11.3 Module safety model

Module commands and custom arguments are tokenized without general shell evaluation. Pipelines, redirection, command substitution, unquoted shell operators, environment-variable expansion, and glob expansion are intentionally restricted in the module format. Review every external module before loading it.

12. Output, Auto-Save, and Command History

When Auto-Save is enabled, supported command output is written beneath the configured output directory using the target and functional category. The default root is **./results**.

```
results/
`-- TARGET/
    |-- recon/
    |-- web/
    |-- network/
    |-- smb/
    |-- traffic/
    |-- external/
    `-- ...
```

Tool_Box also records command history in memory and, when possible, persists it under the output root as **tool_box_command_history.log**. Result entries record target, exit status, and saved output path when available.

Sensitive output

Tool_Box redacts selected sensitive arguments in its own history, but saved third-party command output can still contain usernames, addresses, credentials, findings, or other sensitive information. Review files before publishing or committing them.

13. Configuration and Local State

Common per-user Tool_Box files include:

```
~/.tool_box.conf  
~/.tool_box.modules.conf  
~/.tool_box_terms.conf  
~/.tool_box_profiles/
```

- **~/.tool_box.conf** - saved default threads, wordlist, output directory, and mode toggles.
- **~/.tool_box.modules.conf** - selected external-module filenames.
- **~/.tool_box_terms.conf** - terms revision, Tool_Box version, and acceptance timestamp.
- **~/.tool_box_profiles/** - reusable settings profiles.

Standard-user and root sessions can have different per-user configuration/module-selection files.

14. Recommended Workflow

- 1 **Create or load a Workspace** under Target & Scanning.
- 2 **Set Target / Data:** IP/subnet, ports, output folder, and any domain/URL/user values you will reuse.
- 3 **Enable Dry-Run** while testing unfamiliar command builders or modules.
- 4 **Run Nmap or a Scan Profile** to establish basic host/service information.
- 5 **Import Nmap Results** into the active workspace when you have saved output.
- 6 **Review Service-Aware Recommendations** to choose the relevant enumeration menu.
- 7 **Record Notes and Findings** instead of relying only on raw tool output.
- 8 **Use Results / Reporting** to search, compare, archive, and generate a Markdown target report.

15. Troubleshooting

A tool says it is missing

Open the relevant Software page or System & Configuration -> Install / Manage Software. Confirm the executable is installed and in PATH.

A command needs root

Use the privilege guidance shown by Tool_Box. You can start a nested root Tool_Box under System & Configuration -> Privilege / User Session.

Results are not saving

Check Settings -> Output Directory and Auto-Save Output, then run Diagnostics -> Output Folder Check.

A module does not appear

Open Settings -> External Modules, confirm it is discovered/selected, then rescan/reload. Use the separate External Module Guide for module-format troubleshooting.

A module placeholder keeps prompting

The global value may be empty. Use C - Custom Placeholder Values on that module, or populate the corresponding shared Target/Data field.

Documentation download fails

Check Internet/DNS access, the configured repository path, download directory permissions, and Diagnostics network/repository checks.

Color output looks wrong

Disable Color Output in Settings or launch with a terminal/environment that supports ANSI sequences.

Saved history contains unexpected information

Review the third-party program behavior and result files. Tool_Box redaction only covers values it explicitly handles.

16. Security, Privacy, and Responsible Use

- Use Tool_Box only on systems and data you own or are explicitly authorized to test.
- Review Command Preview before important or intrusive actions.
- Use Dry-Run while testing new workflows or external modules.
- Avoid running the entire tool as root unless necessary.
- Do not hard-code real passwords, API tokens, private keys, or other secrets into module files.
- Remember that third-party programs can display, log, or transmit sensitive values even when Tool_Box redacts its own preview/history.
- Back up important data and use isolated lab networks for testing intrusive tools.
- Review saved results before publishing or pushing them to GitHub.

License and warranty

Tool_Box is distributed under the MIT License and is provided as-is, without warranty. Read LICENSE and TERMS.md in the repository for the project notice and authorized-use acknowledgement.

17. Quick Reference

Item	Reference
Run command	R
Back	0
Main Menu	M
Set Target/Data	T
Workspace	W
Quick Help	H
Default output root	./results
Saved configuration	~/tool_box.conf
Module selection	~/tool_box.modules.conf
Terms acceptance	~/tool_box_terms.conf
Settings profiles	~/tool_box_profiles/
Full guide	Tool_Box_Functionality_Guide.pdf
Quick guide	Tool_Box_Quick_Guide.txt
Module creation guide	Tool_Box_External_Module_Guide.pdf

Appendix A. Hierarchical Menu Reference

Target & Scanning

- Workspace / Project
- Set Target / Data
- Scan Profiles
- Import Nmap Results
- Service-Aware Recommendations

Reconnaissance & Enumeration

- Reconnaissance -> Nmap, Host Discovery, Port Discovery, Service Detection, DNS, WHOIS, Traceroute, Ping
- Web Enumeration -> Gobuster, FFUF, Feroxbuster, WhatWeb, Nikto, Curl headers, TLS inspection, HTTP/HTTPS test
- Network / DNS -> DNS, reverse DNS, WHOIS, traceroute, ARP scan, Netcat test, local interfaces, routing table
- SMB / Windows -> enum4linux-ng, share listing, SMB client, Nmap SMB scripts, NetBIOS
- Service Enumeration -> SSH, FTP, SMTP, SNMP, LDAP, NFS, RPC, database services
- Vulnerability Assessment -> selected Nmap/NSE/TLS/HTTP/SMB assessment checks

Access & Exploitation

- Pass / Hash Cracking -> hash identification and text/encoding tools
- Exploit -> Metasploit

Analysis & Reporting

- Traffic Analysis -> tcpdump capture/read/interfaces/filters and TShark analysis
- Results / Reporting -> dashboards, browse/search/compare, reports, archive/delete, history

Utilities & Resources

- Command-Line Utilities -> Curl, Wget, Netcat, JQ, OpenSSL
- Network Information / VPN Status
- Text / Encoding Tools
- Port Reference
- Useful Links -> shell/session, privilege references, data/analysis, web security, recon/vulnerability, payloads/wordlists, training, custom bookmarks
- Quick Help / Workflow Guide

System & Configuration

- Install / Manage Software
- Settings -> defaults, modes, configuration, profiles, external modules
- Tool_Box Diagnostics
- Privilege / User Session
- About / Version / Update Check
- Documentation / Download Guides

Module-writing documentation

For step-by-step external-module creation, reserved option markers, category aliases, templates, and validation examples, use **Tool_Box_External_Module_Guide.pdf**.